PBL: Utility of Explainability

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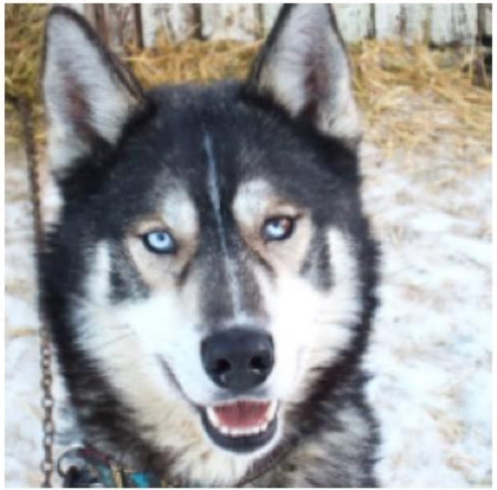
Is Transparency Useful???

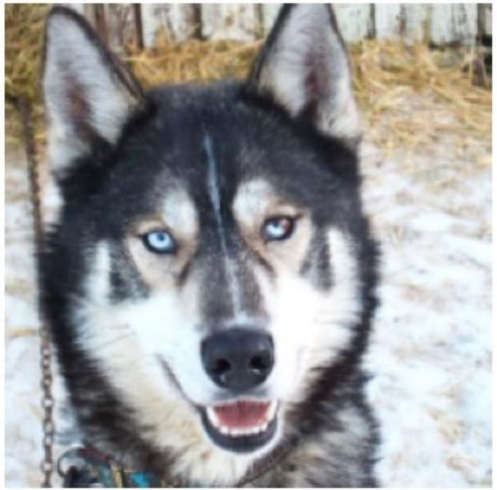
Allows Scrutiny of Models !!!

- **Why is this a wolf?**



Allows Scrutiny of Models !!!

- **Why is this a wolf?**

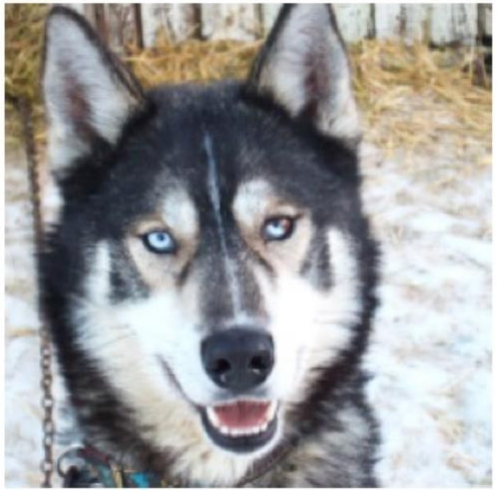


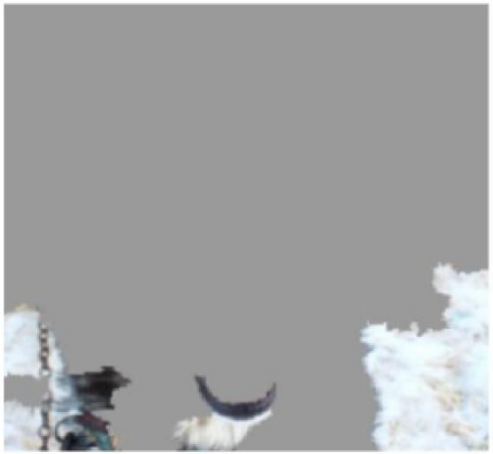
- **It is NOT a wolf**
- **It’s a husky**

https://arxiv.org/abs/1602.04938

Allows Scrutiny of Models !!!

- **Why is this a wolf?**





(b) Explanation

Identifying Vulnerabilities of Models

Human

STOP SIGN

Identifying Vulnerabilities of Models

Model

https://arxiv.org/abs/1707.08945

Is Transparency Hard???

YES !!!

- **There is no widely accepted definition:**

No agreed definition → No standard way to measure

- Competing ideas of what is needed
- More of a philosophical pursuit at the minute than a technical one

What do we need from an explanation?

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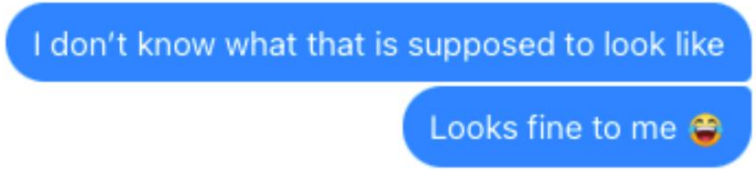
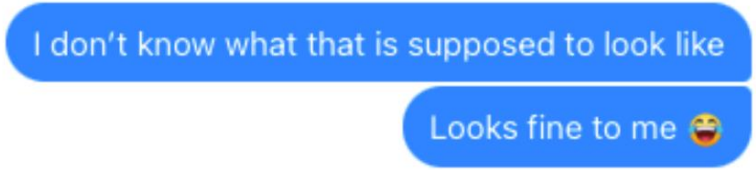


What do we need from an explanation?

Human Explanation



What do we need from an explanation?

You can see a fracture

And the knuckle but should be more central to the bone

What do we need from an explanation?

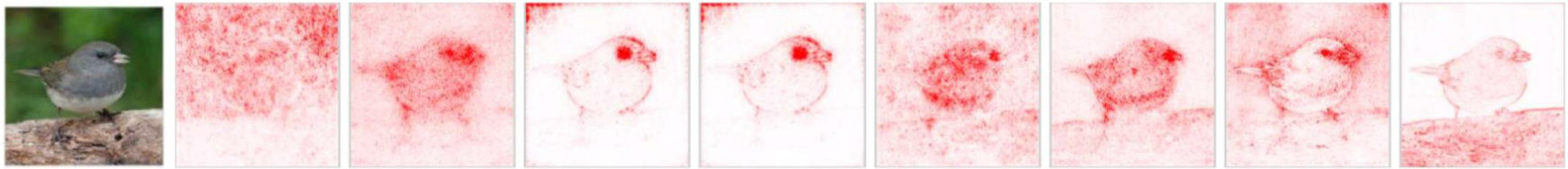
Do humans rationalise before acting? Or vice versa

You can see a fracture

And the knuckle but should be more central to the bone

What is a good explanation?

Junco Bird



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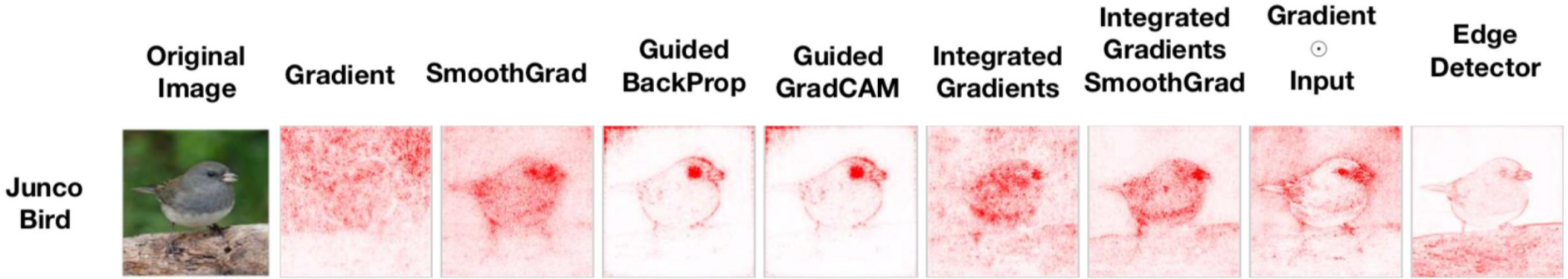
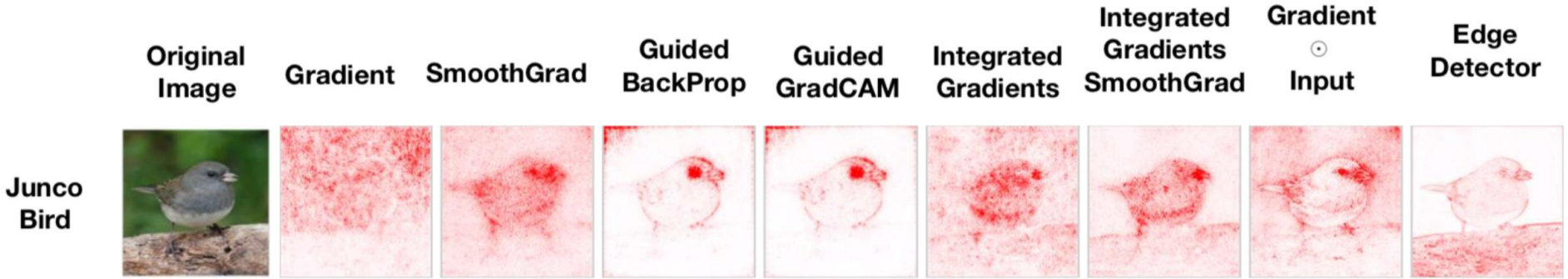


Are Explanations Always Correct???

What is a good explanation?



Do explainers agree?

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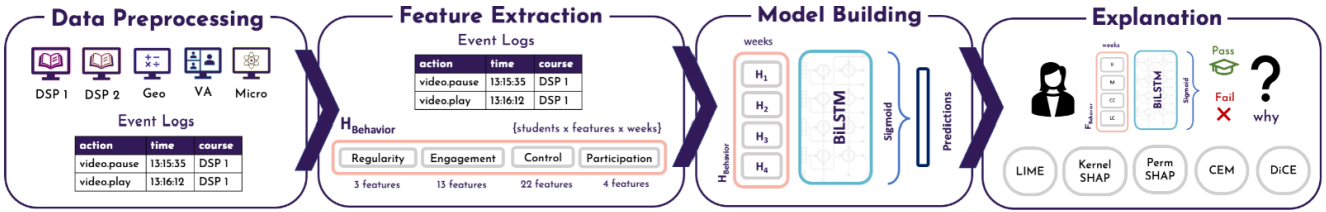
https://educationaldatamining.org/edm2022/proceedings/2022.EDM-long-papers.9/

Evaluating the Explainers: Black-Box Explainable Machine Learning for Student Success Prediction in MOOCs

Vinitra Swamy, EPFL
Bahar Radmehr, Sharif University
Natasa Krco, EPFL
Mirko Marras, University of Cagliari
Tanja Käser, EPFL



- 5 distinct MOOCs
- 5 Explanation techniques

Do explainers agree?

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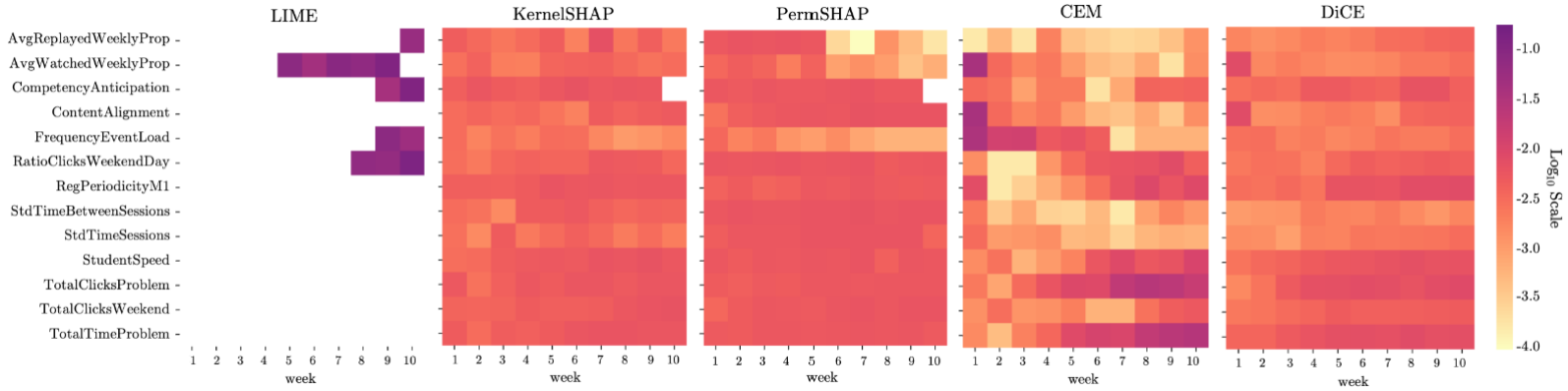

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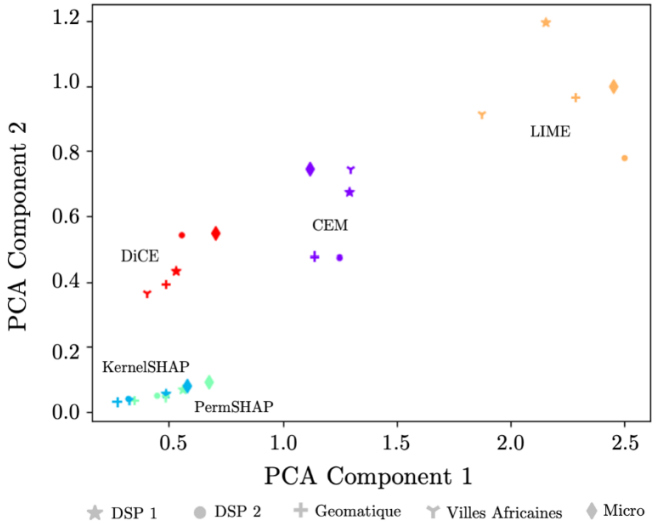

Thank You

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